

DEMO SCRIPT

Show at least:

1. One normal happy-path demo
2. One failure of a naïve approach
3. One advanced technique fixing the failure
4. One edge case where the system should refuse, lower confidence, or ask for human review
5. One trace or explanation proving why the system behaved the way it did

The best teams should not merely show that the system gives an answer.

They should show **why their advanced architecture improves quality, safety, reliability, and trust** compared with a naïve baseline.

SAMPLE PROJECT 5 DEMOS

Golden Scenario 1: Brand-to-Generic Resolution Enables Pharmacy Matching

Technique highlighted

- Semantic medication normalization
- Brand-to-generic mapping
- Cross-system terminology resolution

EHR medication

```
{  
  "brand_name": "Augmentin",  
  "dose": "625 mg",  
  "frequency": "BD",  
  "duration": "5 days"  
}
```

Pharmacy inventory

```
{  
  "generic_name": "Amoxicillin + Clavulanate",  
  "strength": "625 mg",
```

```
"stock": 20
```

```
}
```

Naïve behavior

Bad result:

Augmentin not found in pharmacy inventory.

Advanced behavior

Good result:

```
{
```

```
  "ehr_medication": "Augmentin 625 mg",
```

```
  "resolved_generic": "Amoxicillin + Clavulanate 625 mg",
```

```
  "stock_available": true,
```

```
  "quantity_required": 10
```

```
}
```

What this proves

Hospital systems use different vocabularies. Semantic and dictionary-based normalization reduces discharge delays.

Golden Scenario 2: Out-of-Stock Drug with Alternative Suggestion

Technique highlighted

- Pharmacy stock checking
- Alternative recommendation
- Human approval flag

EHR medication

```
{
```

```
  "brand_name": "Augmentin",
```

```
  "dose": "625 mg",
```

```
  "frequency": "BD",
```

```
  "duration": "5 days"
```

```
}
```

Pharmacy inventory

```
{  
  "generic_name": "Amoxicillin + Clavulanate 625 mg",  
  "stock": 0,  
  "alternatives": [  
    {  
      "generic_name": "Amoxicillin + Clavulanate 1 g",  
      "stock": 15,  
      "requires_doctor_approval": true  
    }  
  ]  
}
```

Expected output

```
{  
  "stock_status": "out_of_stock",  
  "suggested_alternative": "Amoxicillin + Clavulanate 1 g",  
  "requires_doctor_approval": true,  
  "workflow_status": "awaiting_clinical_approval"  
}
```

What this proves

Automation should not silently substitute medications. It should flag alternatives for clinical approval.

Golden Scenario 3: RBAC Blocks Billing from Clinical Notes

Technique highlighted

- Tool-level RBAC
- PHI boundary enforcement

- MCP protocol-level access control

Billing agent request

Call EHR tool: `get_clinical_notes(patient_id=P-1007)`

Expected response

```
{  
  "status": "access_denied",  
  "reason": "Billing role is not authorized to access clinical notes",  
  "allowed_tools": ["create_invoice", "validate_invoice", "get_billing_code"]  
}
```

Naïve behavior

Bad system:

Billing receives full clinical discharge note.

Advanced behavior

Good system:

Billing receives only billable medication line items and invoice-safe data.

What this proves

PHI safety should be enforced by tool permissions, not just prompt instructions.

Golden Scenario 4: Allergy-Safe Alternative Blocking

Technique highlighted

- Cross-tool reasoning
- EHR allergy check
- Pharmacy alternative safety

EHR allergy data

```
{  
  "patient_id": "P-2001",  
  "allergies": ["penicillin"]  
}
```

Prescribed medication

```
{  
  "brand_name": "Augmentin",  
  "generic": "Amoxicillin + Clavulanate"  
}
```

Pharmacy suggestion

```
{  
  "alternative": "Amoxicillin + Clavulanate 1 g"  
}
```

Expected behavior

```
{  
  "alternative_status": "blocked",  
  "reason": "Patient has penicillin allergy. Amoxicillin-containing alternative is unsafe.",  
  "requires_doctor_review": true  
}
```

What this proves

The discharge agent must coordinate across EHR and pharmacy safely, not just optimize stock availability.

Golden Scenario 5: Duplicate Billing Prevention

Technique highlighted

- Invoice validation
- Medication normalization
- Billing safety

EHR medication list

```
[  
  {  
    "brand_name": "Crocic",
```

```
"dose": "500 mg",
"frequency": "TDS",
"duration": "3 days"
},
{
  "generic_name": "Paracetamol",
  "dose": "500 mg",
  "frequency": "TDS",
  "duration": "3 days"
}
]
```

Naïve invoice

```
{
  "billable_items": [
    "Crocine 500 mg quantity 9",
    "Paracetamol 500 mg quantity 9"
  ]
}
```

This double bills the same medication.

Advanced behavior

```
{
  "duplicate_detected": true,
  "normalized_medication": "Paracetamol 500 mg",
  "final_quantity": 9,
  "removed_duplicate": "Crocine 500 mg"
}
```

What this proves

Medication normalization is needed not only for pharmacy matching, but also for accurate billing.